WRITTEN BY A STUDENT, FOR STUDENTS

ENGINEERING + PHYSICS

UNDERSTAND THE REAL WORLD



STEM. 3
ENGINEERING

What is STEM.3?

STEM.3 is a support book, designed and tailored to support KS3 learners in Wales, in particular those with limited access to free knowledge and understanding. STEM.3 provides interactive labs and experiments online. Content is intentionally simplified, and aligned to core KS3 science concepts.

Welcome to STEM.3, and come along on the journey to reach 1,000 students.

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****NOTE** All activities are intended for educational use only and must be carried out under appropriate adult supervision. Only household materials should be used. Schools and guardians are responsible for ensuring activities comply with their own safety policies.**

PART 1: CORE CONCEPTS IN MOTION

1. Newton's Blueprint - Laws of Motion

Newton's laws of motion are three "rules" that explain the relationship between motion of an object and the forces acting on it.

The laws are below:

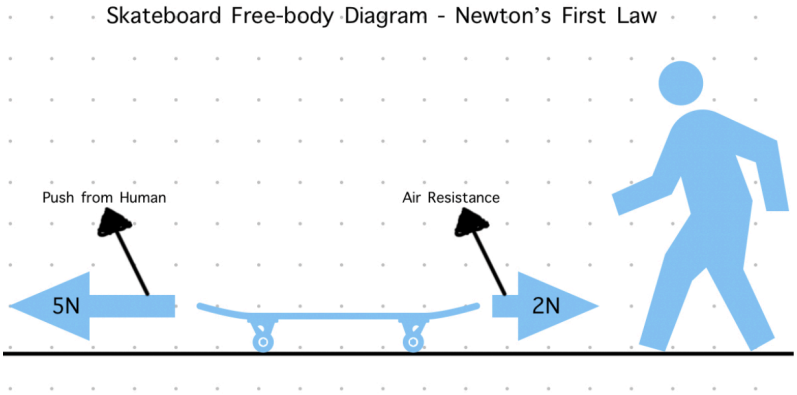
1. *"An object at rest stays at rest, and an object moving at a constant velocity (speed) continues moving at that velocity, unless a resultant force acts on it."*
2. *"At any time, the resultant (total) force is equal to the acceleration of the object multiplied by its mass."*

   Force = mass x acceleration (F = ma)

3. *"Every action has an opposite force acting on the object."*

These three laws may be hard to grasp. Here are some real life examples below:

Newton's First Law:

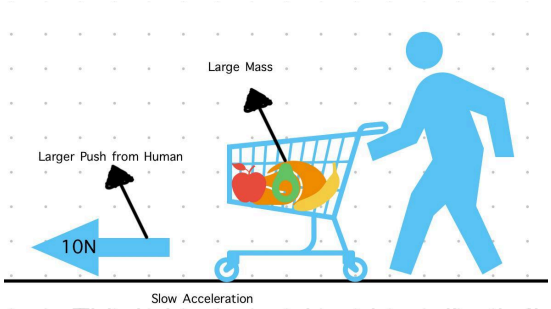


For example, the first law can be explained as imagining a skateboard (left).

Once you push off the skateboard, eventually the air resistance and friction would slow it down. But, Newton's First Law says that if there wasn't any *friction or air resistance*, the skateboard would keep moving at a constant speed. If we remove the 2N of air resistance, the object will be moving forwards at 5N. This is because no force is interacting with the object to stop it.

Newton's Second Law:

Trolley Trouble - Newton's Second Law

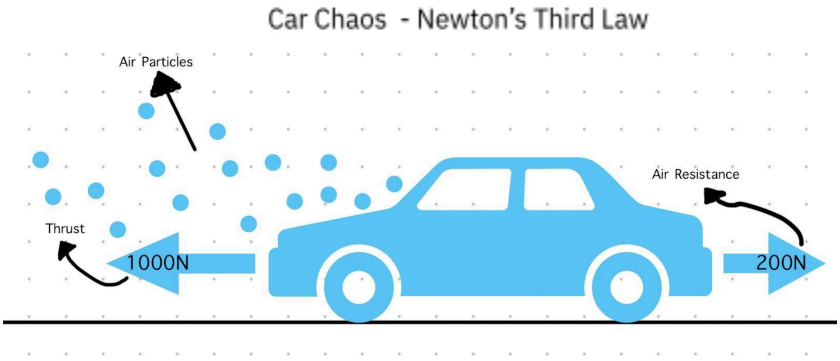


Imagine you're at a grocery store, and you are pushing an empty trolley like a racing car. It will easily speed up with the slightest amount of force. But, as you progress through the market, putting more and more things inside the trolley, you need more force to push the trolley to the same speed as before.

This is because when the trolley is light and empty, the mass is extremely small, but the acceleration is big. However, when the trolley is heavy, mass is extremely big due to the objects in the trolley. To get the same amount of acceleration you need to apply more force.

This shows that acceleration relies on force and mass.

Newton's Third Law:



Let us imagine a car, and as you push the acceleration, the car's speed increases. This increases thrust, but increases drag. Note: drag (air resistance) is not the reaction force to thrust, but is a separate force from the air acting on the car

This is because as thrust increases, the car is cutting through more air but the amount of particles on the car increases, which increases air resistance. This is why if you take your foot off the gas, you will slow down!

This explains Newton's third law, as the action and reaction forces always act on different objects, and counteract each other.

Have your own go!

Blow up a balloon and let go, once it has been let go, you will feel a rush of air hitting your face as the balloon flies off. This is because the balloon forward force causes an opposite force. A prime example of Newton's Third Law!

Create a diagram showing your shopping and label all the forces! If you really want a challenge, calculate the acceleration with this formula: $A = F/m$

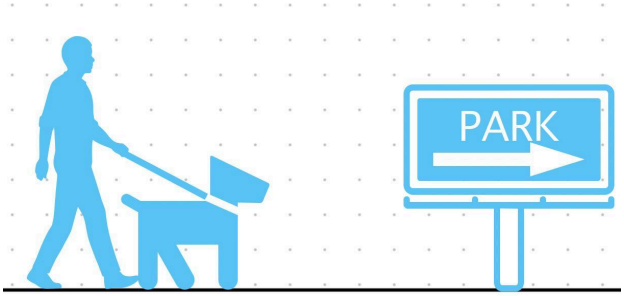
2. Energy: The Invisible Currency

Energy is the source of everything. Your fan, lights, kitchen, and even your body are powered by energy. But what is energy?

Well, energy comes in many different forms:

- Kinetic
- Potential
- Thermal
- And many more.

All these types of energy are transferred in different ways.



Kinetic Energy:

Meet Jerry. Jerry is walking his dog , Bob, to the park. Both Jerry and Bob have kinetic energy; however, Jerry has greater kinetic energy because he has more mass than Bob, even though they are walking at the same speed.

Kinetic energy is possessed when an object is moving (in motion).



Potential Energy:

Jerry has finished hiking a gargantuan mountain accompanied by Bob. They reached the top and adored the view. But even though they aren't moving, they possess gravitational potential energy - a type of potential energy.

This is because potential energy is energy that is stored within an object due to its position, shape, or condition. In this example, it's the position because farther away you are from the ground, the greater the gravitational potential energy you have!



Thermal Energy:

Jerry and Bob are at the top of the mountain, but it's extremely cold, so they started a fire. This fire emits thermal energy as it gives off heat. This heat radiates into its surroundings, making the area warmer.

This energy is created because of the movement of the particles inside the matter.

This is because the warmer the object is, the greater the energy there is because it causes the particles to vibrate quicker.

3. Forces & Friction: The Push and Pull of the World

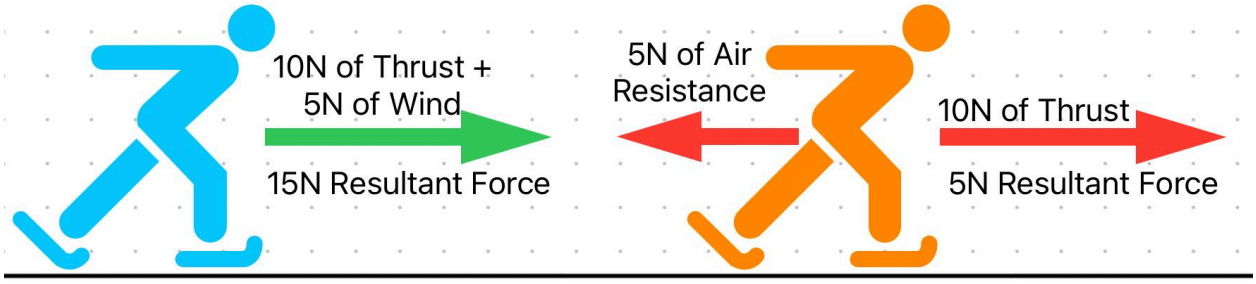
Forces:

A force is push or pull.

Force has something called a ***resultant force*** , and this is the overall force acting on an object. Basically the one force that summarizes all the forces acting on the object.

If the two forces are acting in the same direction, like the wind blowing you forward and you pushing off your ice skate, the wind force and the pushing force will be added up to show the overall force.

However, if the two forces are in the opposite direction, the resultant force is the difference of the two forces in the direction of the larger force. An example of this is when you drive a car! When you press the accelerator, your forward force/thrust needs to be bigger than the force of air resistance. That's why you need to continuously press the accelerator, or you will slow down and stop.



If the resultant force is zero, the object will move at a constant speed or be stationary. This is like if you accelerate extremely hard during the launching phase of driving a car, but after 10 minutes the wind picks up and the wind resistance is equal to the thrust, you will move at a constant speed because you had that initial acceleration before.

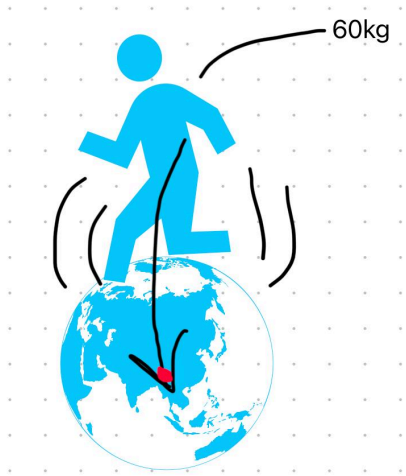
But, if the resultant force on an object is not zero, the object will accelerate in the direction of the larger force (resultant force)

Gravity:

Gravity is what keeps us all together and on the floor. Gravity is the force that draws an object to an object. Gravity keeps us in orbit around the Sun! The same gravity that keeps us on Earth, also keeps us orbiting the Sun, and the Moon orbiting us! *The same gravity that keeps you on Earth also keeps Earth orbiting the Sun, and the Moon orbiting us*

What does gravity do:

Meet Johnny. Johnny lands on the ground when he jumps, but why is that?



This is gravity acting! Earth's gravity is what keeps you and the ground and what makes things fall. This is because Earth's gravity is pulling objects towards its centre, and since Earth is a sphere, when you jump, everything lands on the surface, as if it's being pulled towards the centre.

Anything that has mass has gravity, even a ping pong ball! However, gravity is only noticeable with large objects. That's why the Moon orbits Earth instead of Earth orbiting the Moon!

Gravity also gets weaker with distance and less mass. So the closer you are to an object, the stronger the gravitational force between each other. More mass = More gravity

Earth's gravity comes from its mass, all its mass and gravitational pull gives your weight. Gravity is also a prime example of Newton's third law, as you pull on the Earth, but it's so big that your pull doesn't move it!

Your mass doesn't change depending on what object you're on, but your mass does. This is because mass is the amount of matter (stuff) in an object, and is measured in kilograms, kg. It doesn't change with location. However, weight is mass * gravitational force.

If you go to any other planet, your mass will be if you weight 70kg:

Planet	Mass (kg)
Mercury	25.9
Venus	62.1
Earth	70.0
Moon	11.3
Mars	26.0
Jupiter	173.5
Saturn	74.1
Uranus	60.8
Neptune	78.1
Pluto	4.3

We also get tides because of the Moon! This is due to the Moon's gravity also pulling on the Earth's oceans, causing tides.

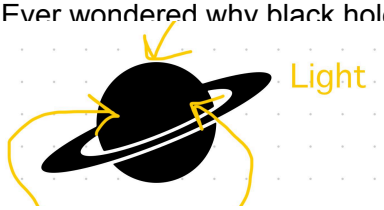
Gravity is also what builds our universe! The heavy materials like rock, are pulled towards the Sun, so all the planets that are fairly close to the sun, are made of ROCK. This is also why planets further away from the Sun are more gassy.

In space, gravity pulls everything together to form an object. They pull things like dust and gas, to form planets, moons, and even stars!

Basically, your mass stays the same no matter where you are, but your weight depends on the gravity of the place you're standing on.

Black Holes:

Ever wondered why black holes are black?



This is because even light is affected by gravity. It bends when it passes near big objects, like black holes. The "white ring" around a black hole is the light bending around

it. No light ever escapes the black hole. This is what Einstein described gravity is.

In summary in black holes, gravity is so strong that not even light can escape it, so it forms rings around it.

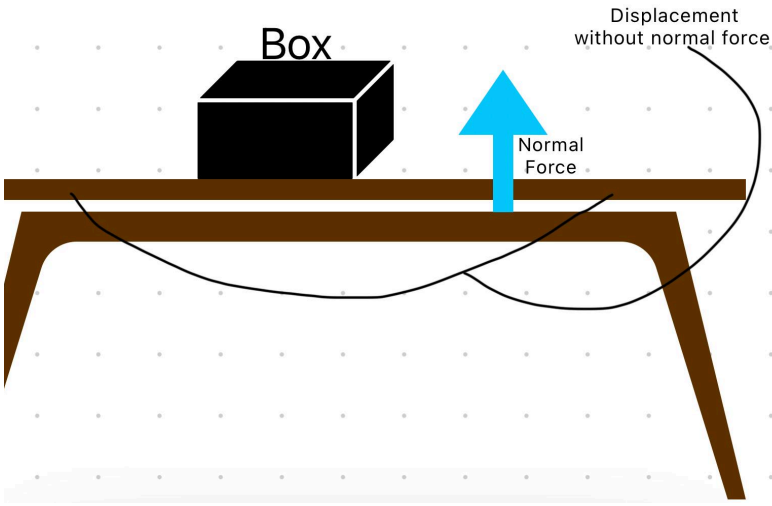
Fun Fact:

Earth's gravity is a bit stronger in places that have more mass underneath it.

Normal Force:

"Normal force is a support force that is exerted by a surface upon an object, acting perpendicular (90˚) to the surface of contact."

It's not called normal because it's ordinary, but because it is perpendicular to the surface.



Example:

When an object, like a box, sits on a table, the table pushes back up on the box, to support the box from breaking through. This upward push is called the normal force.

If this force wasn't there, the object (the box) would fall straight through due to gravity! The object would accelerate downward to the fall.

The balance between these two forces match the force trying to press into the surface (table). That force is regularly the weight. This is also an example of Newton's Third Law!

The normal force always acts at a right angle to the surface.

No exception. It also only exists when two surfaces are touching. If the weight of the object is quite big, the deformation of the surface is greater, so the normal force is larger. This is kind of like a spring pushing back.

How Does the Table "Know"?

Great question! But the table isn't alive, so how does it know when an object is on it?

That's because the table acts like a spring, and when an object is placed on it, it becomes a compressed spring and pushes the object upwards, providing the normal force. This restoring force increases until it balances the object's weight.

In a nutshell:

I know it may sound hard, but normal force is just reactionary; it adjusts based on what's pressing into the surface.

It always acts perpendicular to the surface.

Even inanimate objects and surfaces like tables exert forces due to deformation and elasticity, kind of like spring!

Friction:

Friction is the pull of the world. It's a force between two surfaces sliding, or trying to slide across each other. Like pushing a toy car on carpet, but compared to pushing it on a table, friction makes it harder!

Friction always works in the opposite direction of the movement. Like Newton's Third Law! It just slows things down. The amount of friction is dependent on how rough your surface is. The rougher the surface, like carpet, the more friction.

Friction also produces heat and thermal energy. That's why when you rub your hands quickly, they warm up!



Why is Friction Useful?

Friction may seem overall negative, but it has some positives. For example, Friction is useful because it stops your shoes from slipping, and planes landing as the friction slows down the plane, improving fuel efficiency,

But, we may want to reduce friction, like oiling the gears on a bike. The oil keeps parts apart so they slide smoothly, reducing wear.

Tension:

Tension is a pulling or stretching force that is passed through an object like a rope, string, chain, or rod.



Example:

Nine men are pulling a rope. Each small segment of the rope feels a force pulling it from both sides. This pulling force is called **tension**. This force acts along the length of the rope - this is why each part of the rope feels the same tensity.

What causes tension?

Tension occurs when an object is being stretched or pulled, so the object in that example is the rope. This tension can only happen because of the atoms and molecules being pulled apart.

The atoms are connected by forces which hold themselves together. When you stretch these atoms, the forces holding them together are pulled apart from the regular positions. This stores potential energy in the bonds which are stretched within the atom.

This is similar to a spring. The forces between the atoms are trying to pull back to their original positions, so they exert a restoring force, vaguely similar to normal force. However, this restoring force is counteracted by the stretching of the rope, making the rope tensed.

Think of it like an elastic band, when you stretch it, it wants to return to its original position, but it can’t due to your stretching, so it feels tense.

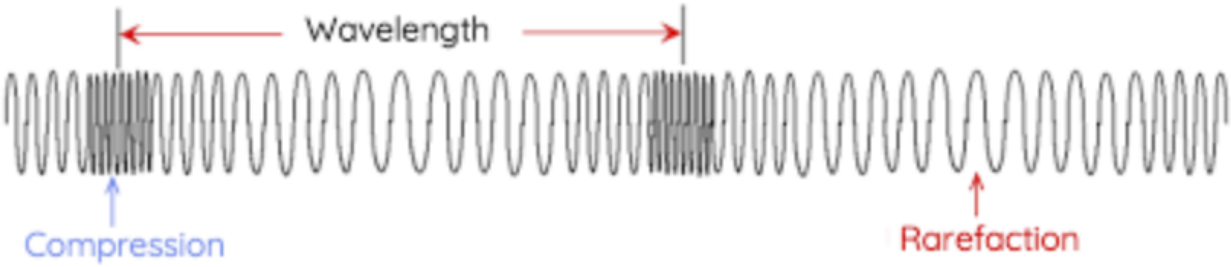
PART 2: WAVES, SOUND, LIGHT & ELECTRICITY

Sound Waves: Physics You Can Hear

Sound waves are the things that help us, homo sapiens, conquer the world. Okay, but let’s be serious, sound is a mechanical wave. This means it only works in a medium (air, liquid, or solid) to travel. That’s why you can’t talk in space, because it’s a vacuum, sound cannot travel so you can’t hear each other in space.

Sound waves are longitudinal pressure waves, which means they have compressions and rarefactions parallel to the direction of energy transfer.

Sound needs something to vibrate, if there is no air in space it means no sound. Astronauts talk via radios because the radio signal travels through space however.



The compressions are the high-pressure regions, and the rarefactions are low-pressure regions. These longitudinal waves move side to side, which gives the pressure differences.

The key properties of sound waves include:

- **Amplitude** – reflects the size of the pressure differences which determines loudness - higher amplitude = louder sound.
- **Frequency** – number of wavelengths cycled per second measured in Hertz. It relates to the pitch - higher frequency = higher pitch.
- **Wavelength** – the length of one complete wave cycle.
- **Compressions** - areas of high pressure, where the oscillations are close together.
- **Rarefactions** - areas of low pressure where the oscillations are far apart.

The speed of sound changes, as it depends on how dense the medium is. This is because the waves can travel in denser mediums. For example, sound travels fastest in solids, slower in liquids, and slowest in gases due to the differences in density.

Everyday and Advanced Applications

Sound waves are an essential part of the real world apart from speaking, listening, etc.

For example, echolocation is a prime example of the use of sound waves. Echolation is a sonar system used by animals. The animals emit high-frequency sound waves into their environment, and then wait and listen for the echoes that bounce back after hitting objects.
By analysing these echoes, they can create a map of their surroundings and navigate through the forests in the complete darkness or in murky water.

These sound waves produced by the animals like bats and dolphins are completely undetectable by humans. This is because they have a frequency higher than what humans can hear, above 20,000 Hz.

The waves travel through the medium (air or water) as mechanical vibrations, so that when they reach back to the animal, the strength of the wave tells them how far away that object is.

By measuring the time delay between the emission and reception, the animal can understand how far away the object is:

$$Distance = \frac{Speed\ of\ sound \times Time\ delay}{2}$$

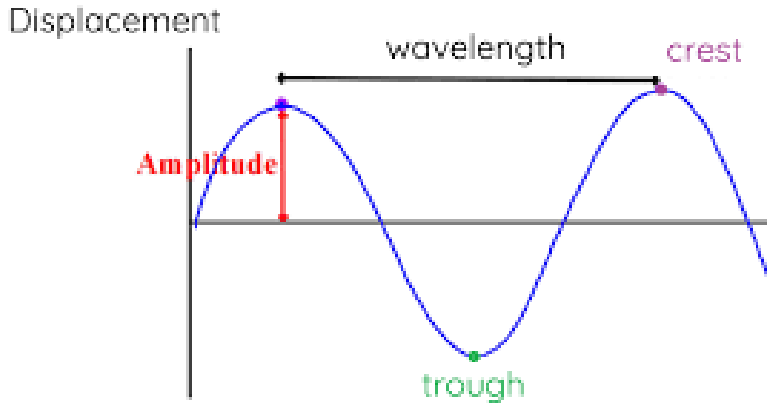
The differences in the echoed waves' intensity, frequency, and timing provides information about the object size, shape, texture, and even movement.

These waves are in a higher frequency because since higher frequency = shorter wavelengths, they allow for a clearer, finer resolution when detecting objects (allows them to detect minuscule insects and objects).

Light: The Bending Truth

Light is electromagnetic radiation of any wavelength, whether it's visible or not.

Light is a **transverse** electromagnetic wave, the other type of wave other than a longitudinal wave (sound). This means the oscillations (vibrations) undulate (move up and down) about the equilibrium position. Their oscillations are perpendicular to the energy of transfer



The highest point of the wave is called a crest or a peak. The lowest point of the wave is called a trough. A wavelength is measured from either one crest to another, OR, one trough to another.

The key properties of light waves include:

- **Amplitude** - the maximum displacement from the equilibrium position. This defines the wave's brightness or intensity – higher amplitude = brighter light.
- **Frequency** - the number of wave cycles per second, measured in Hertz (Hz). For visible light, frequency determines **colour** – higher frequency = violet light, lower frequency = red light. This is why you get a rainbow!
- **Wavelength** - the physical distance between two corresponding points on consecutive waves (e.g., crest to crest). In visible light, shorter wavelengths correspond to blue/violet, and longer wavelengths to red.

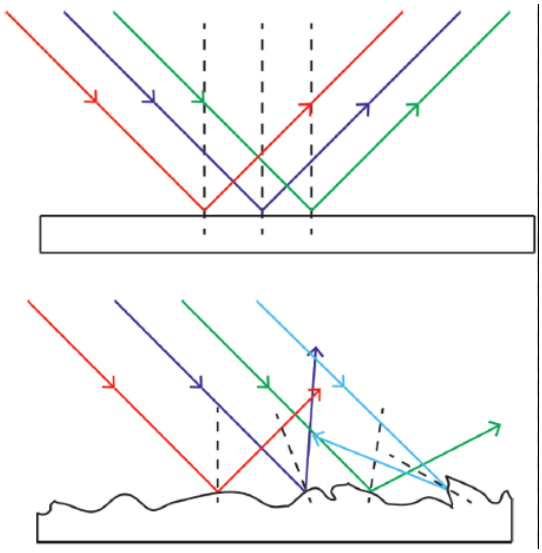
The **speed of light** in a vacuum is about 3 * 10^8 in a vacuum, or 300,000,000 m/s which is the fastest possible speed in nature. Light slows down when it travels through optically denser materials (like glass or water) because of refraction, but the frequency remains the same — only wavelength changes.

Light can be transmitted, reflected, or absorbed.

Reflection:
When light is reflected, it follows the law of reflection: angle of incidence = angle of reflection.

The angles of incidence and reflection are measured from the normal which is at 90˚ to

the mirror line.



There are two types of reflection:

- Specular reflection – when light hits a flat and smooth surface like a mirror, and the light reflects in uniform which leads to a clear image being formed.

- Diffuse reflection – when light hits a rough, uneven surface and leads to no image being formed due to the reflected rays being scattered in different directions and no clear image being formed.

Refraction:

Refraction is the bending of these light rays as they change from different mediums. It explains how rainbows are formed as well.

Refraction only occurs when light changes medium at an angle other than the normal line. When light goes from a less optically dense medium, like air, to a more optically dense medium, like water, it bends towards the normal. And vice versa, when light goes from a more optically dense medium to a less optically dense medium it bends away from the normal.

Refraction only happens because light travels more slowly in a more optically dense medium.

Imagine a shopping cart, and this shopping cart is being pushed from a smooth floor to thick carpet by the world-famous BOB!

When Bob is pushing the cart on a smooth tile floor, the cart rolls quickly. But when Bob pushes it at an angle, one wheel hits the carpet first, so that wheel slows down first. This means that the other wheel is still on the smooth floor, moving faster.

This overall increases the speed on the left side of the cart, making it turn right towards the normal.

How rainbows are formed:

Rainbows are only formed when the sun is present and the rain as well. This is because each raindrop acts as prisms, and when light travels through these prisms, they refract. This light is white light which contains all the colours of the rainbow.

The process of rainbow formation includes a combination of refraction, internal reflection, and dispersion within the teardrop.

When the white light is refracted, a rainbow is formed due to the different colours getting refracted by different amounts. They get refracted by different amounts due to their frequency.

For example, light with the highest frequency (violet) bends the most, and the light with the lowest frequency (red) bends the least. That's why you get red at the top – with the longest gradual bend – and violet at the bottom – the shortest instant bend. The light gets refracted in, reflected inside, then refracted out again.

How lenses work – A Magnifying Glass?

Lenses are probably one of the most interesting forms of refraction. This is because a lens is only a transparent material that is curved and bends light. But how does the light bend to focus on one particular point?

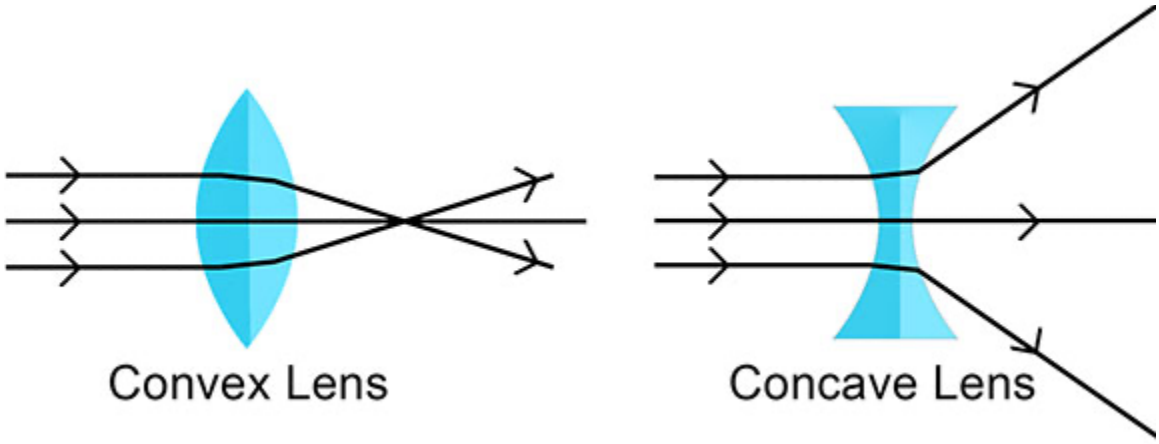
The light bends due to refraction. That's because the light is passing through a different material, glass – the lens.

Did you know that each light ray enters the glass at an angle? This change in speed causes it to change direction, and the lens has different levels of curvature which allows each ray that hits the glass at any angle, which refract into the direction of the normal!

These rays eventually come out of the glass which are called the emergent rays. These emerging rays exit the glass into the air, they bend again, BUT they bend away from the normal as air is less optically dense than the previous material, glass.

The effect of this is that the lens is convex, so the light rays bend towards each other and meet at a point which is the focus. This means that your eye sees the object as larger because the lens creates a virtual, upright, and magnified image when the object is within the lens's focal length (distance to object and lens).

The convex lens works because the light bends towards the centre, so it creates a larger virtual image. However, a concave lens bends light away from the centre and spreads out the rays to form a smaller virtual image.



Electricity & Circuits: Wired for Power

What is electricity?

Well, electricity is the flow of charged particles – specifically electrons. Electrons are **negatively charged** particles: -1 charge. It has almost 0 mass, but to be specific 9.109 x 10^{-31} kilograms.

However, **static electricity** is the build up of electrons on an insulator – a material that doesn't allow electricity or heat to pass through it easily.

Did you know?

- Electricity was discovered more than 2500 years ago – the Ancient Greeks first discovered static electricity.
- One lightning bolt contains enough energy to toast about 100,000 slices of bread – that's a lot of cheddar!

The speed of electrical signals is very fast (close to the speed of light), but it depends on the material. However, individual electrons move much slower, so when you flip a switch, the light appears to turn on instantly even though the electrons themselves move slowly. So the message to turn on the light is very fast, but the speed of the electrons is slower.

Current

Electrical current is a flow of charge, and in a conductor (a material that allows electricity to pass through easily) of electricity, the charge is the flow of electrons.

For it to flow, you need two things:

- Something to transfer energy to the electrons – a battery
- A complete path that connects the start and end together for the electrons to flow through – an electric circuit.

You measure current by using an ammeter. The level of current is the constant throughout the circuit, and never changes.

Voltage

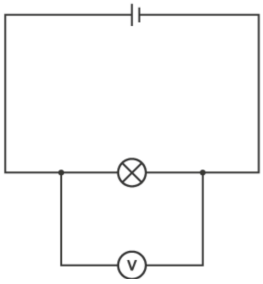
The voltage is a measure of difference in electrical energy between two different parts of a circuit. The bigger the difference, the bigger the voltage.

Voltage is measured in volts (V). The bigger the amount of volts, the bigger the electrical energy.

Fun Fact:

Voltage is also called potential difference.

You can measure voltage using a voltmeter. To use a voltmeter, you must place it in parallel with the circuit:



You can calculate voltage by using an formula:

$$V=IR$$

$$Voltage = Current \times Resistance$$

Resistance

- Resistance is a measure of how difficult it is for current to flow. The lower the resistance, the easier it is for current to flow.
- Resistance is measured in ohms Ω.
- The amount of current flowing in a circuit is affected by the resistance of the circuit.
- Every component in the circuit has some sort of resistance – the lamp, battery, etc.
- As resistance increases, current decreases.

Resistance is caused by the current, which is made of electrons, colliding with ions. When they collide with these ions, it causes resistance. Adding more resistance = adding more ions.

This is why if you use a battery to power two things instead of one, it takes longer since there is more resistance; however, if you were to charge only one thing at a time, it would be faster due to a lower amount of resistance.

Did you know that the symbol for resistance actually means *Omega*?

Even the wires have resistance, but they are so minuscule that they are often ignored.

You can calculate resistance by using an formula:

$$R = V \div I$$

$$Resistance = Voltage \div Current$$

Ohm's Law

Imagine electricity glowing through a wire, like water through a hose.

- **Voltage** is like the water pressure, pushing the flow along.
- **Current** is like the water itself rushing through.
- **Resistance** is like a bend or a kink in the hose that makes it harder for water to get through.

Ohm's Law brings it all together in one simple rule:

$$V = I \times R$$

The voltage across something = the current flowing × its resistance.

It's kind of like this: more pressure (voltage) means more flow (current)... unless the hose is really kinked (resistance), in which case, not much will get through!

Next time you plug in your phone charge, thank Mr. Georg Ohm for making electricity a little less "shocking" to understand.

How do Car Batteries work?

Just think of your car battery as the hearth of the car's electrical system. Without it, your car would just sit there like a massive metal box!

So, here's what happens:

- The battery is packed with chemical energy to supply electricity to start the engine.
- When you start the car, the battery sends out electricity to the starter motor. That motor gives the engine and big shove to get it going, and the alternator generates electricity to power the electrical systems.
- But once the engine's on, it doesn't stop – it also powers the lights, wipers, radio, and even that annoying "seatbelt not on" beep!

Well yes and no. Cars are clever. While the engine runs, the alternator (a tiny electrical generator) turns some of the kinetic energy back into electrical energy. This recharges the battery ready for the next journey. However, the battery will eventually wear out after many years.

In a summary, it's like this:

Chemical energy → Electrical energy → Car components
And
Kinetic energy → Electrical energy → Battery recharge

Next time you're in the car singing along to the radio, think of the battery and the alternator – the ultimate power duo keeping your car alive!

PART 3: THE ENGINEER'S BLUEPRINT

The Design Process: Think, Plan, Build

Engineers don't just create a product or invention in one go – they follow a cycle, and **iterate** over time. This will eventually lead to a better, more sustainable product or invention and real, working innovation. This process is called the **Engineering Design Cycle**.

The main steps are:

1. **Define** the problem – why do you need this invention, what problem does it solve that people face?
2. **Identify** & **research** - figure out what needs to be done: the design requirements and your limitations.
3. **Brainstorm** ideas – what creative ideas can you make – look at other innovations for inspiration? Determine what will be cheaper and faster, and what everyone wants and what **you** have.
4. **Plan** – sketch ideas, choose what materials you're going to build it out of, and make a step-by-step design.
5. Build a **prototype** – build the MVP – the Minimum Viable Product, the easiest version of the invention.
6. **Test** & **Improve** – you test the invention. Does it work as expected? What can you improve? Get others to test it out
7. **Feedback** - you get a variety of feedback, and ask questions on what they like and don't like.
8. **Iterate** – build up on the feedback, and tweak bits of the design.
    a. This is called iterating – making changes based on test and user feedback.
9. **Communicate** – present your solution to others, and see what impact it has and how you can improve it.

This cycle never ends – engineers are constantly improving their product from feedback.

Real-life example

All of this 1234 may be a bit much, so let's break it down with a **real** problem – the Dyson Vacuum …

James Dyson, the founder of Dyson, created the original vacuum using the exact cycle. He was frustrated that every time vacuum the traditional vacuum cleaners clogged. This inspired him to create his own vacuum using the **Engineering Design Cycle:**

- **Problem** – traditional vacuum bags clogged up, meaning the vacuum lost suction making it ineffective.
- **Research** – James Dyson saw this problem, and applied the cyclone principle to use centrifugal force to separate dust and dirt from the airflow.
- Brainstorm – he was inspired by a sawmill of all things, to create a bagless vacuum cleaner by adapting the industrial cyclone technology.
- Testing – this overall led to the point where he developed a multi-stage, multi-cyclone filtration system.
- Prototyping and Iteration – Sir James Dyson made thousands of prototypes, changing each system through tough testing.
- Communicate – he created a vacuum and led to him changing the market and inspired other industries.

This shows that even the best companies and noble people use this system to build their innovation up.

Materials Matter: Why We Don't Build Bridges with Chocolate

Materials, what are they, things that build up society?

Well, materials are a *bit* more than that. Materials are the substances or matter in which physical objects are made. Materials have, or possess, properties like hardness, density, colour, and conductivity. These specific types of properties determine their usefulness in civilisation.

The four main types of properties are:

- Stress
- Strain
- Elasticity
- Toughness

Stress

What is stress?
Stress is the measure of what a material feels, and how much it can take from forces externally – like normal force.
Why does it matter?
Stress in materials is vital as it determines how easily a material will deform and fail. This eventually results in the safety and reliability of materials and structure of a building due to its material.

Stress allows engineers to predict how materials will behave under load, or external forces. That allows them to design parts strong enough for their needed use without causing deformation or other unwanted changes.

Stress is just the ratio of the number of the external forces to the cross sectional area (normally the middle point) of the material.

Strain

What is strain?
Strain is the amount of deformation of a material caused by stress.

Why does strain matter?
It shows how much deformation can occur under stress, which allows engineers to predict how long a component can last, design it for safety, and predict how well a material can perform.

Strain is simply the ratio of change in length to compared to the original length,

Elasticity

An object is said to be elastic if it returns to its original shape after it has been stretched or compressed.
When this elastic object is stretched or compressed, it stores elastic potential energy.

Most objects, like a spring,obey Hooke's law. We know if an object obeys Hooke's Law when the extension is proportional to the force applied to it.

Hooke's law can be stated as F = kx, where F is the force, x is the extension and k is the spring constant.

The spring constant of a spring is a measure of how stiff it is.

Elasticity matters because it allows materials to deform under stress, and then return to its original shape without any permanent damage. This is vital because it ensures that the construction of bridges or things like that are safe and reliable.

Knowing a material's elasticity is crucial in engineering to predict how it will behave and to calculate how much it can withstand to ensure longevity and performance in normal applications.

Toughness

Toughness is how much a material can absorb energy and deform without breaking.

This matters as much as elasticity, strain, and stress – it builds up the material and tells engineers what it can do, and how to improve it and what to do with its usefulness.

Toughness is measured the same as elasticity and strain – how much can it absorb before fracturing?

What are bridges made out of?

Bridges are built by selecting materials like steel and concrete.

Steel has a high stress resistance and strain, whilst also being tough. This allows it to be durable – perfect for holding the weight of concrete and cars.
Concrete has a high stress tolerability, and is extremely tough. These combined allow for highly durable bridges, ensuring safety and longevity.

This allows bridges to constantly carry the stress of cars and trucks, whilst doing it for decades.

Bernoulli's Principle.

Air may be perceived as invisible and nothing, but it is actually something. Air has a mass, a density, and even a mixture of elements.

Air constantly exerts pressure on everyday objects like us, and these can be predicted and used in industries like aviation by using Bernoulli's Principle.

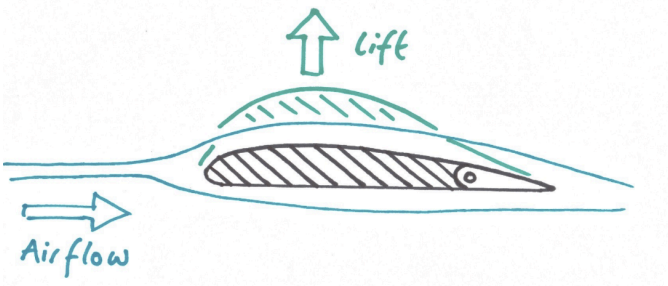
Bernoulli's Principle

Bernoulli's Principle: when a fluid increases in speed, simultaneously either the pressure is decreasing, or the potential energy of that fluid (air) is decreasing at the same time.

This essentially means that as the speed of a fluid increases, the pressure decreases at the same

That principle is the thing that allows you to fly from country to country, as this is the key idea behind how airplanes get off the ground.

This is because when you see a normal wing, it is pretty much straight. That allows the plane to fly through the air with minimal drag. However, when you're taking off, what lift allows the airplane to get off the ground. This is why there are flaps. The flaps are basically an extension of the wing angled downwards.



Flaps do two things:

- Slows down airflow
- Redirects air downwards.

These each have an importance.

Slowing down the airflow allows for an increase in pressure, according to Bernoulli's, and redirecting it downwards means that there is lift generated. This allows the airplane to take off.

These examples show that **controlling airflow and pressure can make things move, fly, or spray efficiently**.

PART 4: BIG BUILDS AND BRAINPOWER

Engineering Structure: Don’t Let It Fall

Compression

Compression is a general term, but it can be used more specifically for structurality. It is the squeezing force that pushes the material’s particles closer. This reduces the space between the particles, reducing the object’s size. It is thus the opposite of tension.

Why is it used?

The forces that push inward on an object, like weight, compact it meaning that the overall structure gets weaker due to its reduced size. This is because it shortens along the force's direction, but slightly widens perpendicular – short and fat (a bit like me after Christmas).

Example:

A building’s support column is used to bear weight, and compression needs to be tested for their compressive strength, which is the maximum force they can take before crushing/deforming. This is why concrete is used, instead of foam, because concrete has a greater compressive strength.

Load Distribution:

It’s how forces spread across a structure, and pressure = force/area is key here.

A force spread over a surface, rather than a single point, can withstand a greater amount of load. This means that a load with a constant magnitude across the entire element, maintains a uniform distribution load.

That also means that load transfer can occur, where loads move from one part of a structure to another. Like a roof to the wall to the foundation, increasing the volume of load.

Load Distribution is vital for stability and strength, as it prevents structural failure by ensuring forces are shared effectively across supporting elements. It’s used vitality in bridges, buildings, and even prosthetics!

Load Distribution = load per unit*length/area.

Aerospace & Rockets: Newton's 3rd Law in Action

The key topics are:

- Thrust
- Drag
- Lift
- Gravity

Thrust

Thrust is the forward force that propels aircrafts in the air by expelling mass. This is related to Newton's Third Law as it directly counteracts drag and is crucial for overcoming weight during takeoff and maintaining speed during the different stages of flight.

It works by accelerating fluid backwards at high speeds, and creates an equal opposite reaction of force, pushing the aircraft forward – Newton's Third Law.

Jet Engines suck in air, compress it, and expel it as hot gas. Propeller Engines push air backwards, pulling aircraft forward. Rockets expel propellant (fuel) at extremely high velocity, generating thrust – in space because it uses propellant, not air.

Stages of Flight in terms of Thrust:

1. Thrust is increased to force lift off and gain altitude
2. Thrust balance to equal drag to maintain constant speed.
3. Increased thrust for acceleration
4. Decreased for deceleration

Thrust is measured in Newtons: Mass Flow Rate of the exhaust and its velocity change.

Drag

Drag is the force opposing the object's motion, through a fluid like air and is caused by direction and pressure differences. If an airplane experiences drag, it causes a resistance, slowing it down, reducing efficiency.

Lift

Lift pushes an object upwards, opposing gravity. It is produced by a fluid moving over a curved surface at different speeds, creating lower pressure above, and higher pressure below, causing it to rise. The pressure differences cause it to rise as the higher pressure underneath overcomes the lower pressure, meaning the resultant force is upwards.

Titling the wing changes the angle of attack, and deflects more air downwards, leading to a greater lift due to Newton's third law pushing the wing upwards.

It counteracts gravity, and acts at a right angle to direction of airflow.

Robotics & Mechatronics

How do Sensors work?

Sensors work by detecting a physical change in the environment, and convert that detection into a n electrical signal, leading to a stimulus like light, heat, or pressure. This signal can be displayed, sent to another device, or used to control an action.

What are Actuators?

Actuators are components that convert energy into a mechanical motion, creating a similar movement like a muscle. It's the device that allows robots to make, and takes control signals and translates it into a physical action, allowing automation in everything – like factories.

Example:

If you use a sensor to detect the temperature of water in a kettle, if it reaches 100˚C, a sensor will convert that direction and send an electrical signal to the actuator to take the kettle off the stove by using arm-like movements to grab and move the kettle.

PART 5: PHYSICS AND THE FUTURE

Renewable Energy: Powering Tomorrow'

There are different types of renewable energy:

- Solar
- Wind
- Hydro

Solar energy is converted into electricity through a process called photovoltaicism. It is where the sunlight strikes semiconductor cells, and knocks electrons to create a direct current. Then, an inverter converts the direct current into alternating current, which is used in homes and organizations.

Wind energy is converted into electricity by using blades to capture the wind's kinetic energy, which spins a shaft connected to a generator. It converts the kinetic mechanical motion into electrical energy, AC power, for homes and the grid.

Hydro energy converts into electricity similar to wind energy, but it uses the water's kinetic energy into electricity by using its force to spin turbines which are connected to generators that

transform this mechanical motion into electrical energy and is used in dams for continuous power sent to transformers for efficient transmission.

Energy transfer is **key**, as it is the relocation of energy from one object, system, or location to another, and usually changes form. However, the total amount of energy is the same, due to the law of energy conservation. It happens via:

- Thermal Work (Heat)
- Mechanical Work (Kinetic/Movement)
- Electrical Work (Wire/Currents)
- Radiation

It's used to power things like phones, as the kinetic energy from the wind is converted into electrical energy, used to power phones.

Engineering Ethics & Environmental Impact

Sustainability is always key in engineering, as you need to understand to ensure your idea is eco-friendly, socially accessible and accepted, and is economically sufficient.

Green Design:
Green Design is an approach to create products, buildings, and systems that minimize the negative environmental impact by reducing pollution, conserving and saving resources, and promoting ecological health throughout the project's idea.

It involves:

- Environmental considerations
- Social considerations
- Economic considerations
- Benefits planet
- Benefits people

It is used to produce renewable energy and recycled materials.

Engineering for Good:

Engineering for Good combines engineering, and humanitarian issues. It is used to apply engineering principles and techniques, to solve social and humanitarian challenges, aiming to create accessible, affordable, and sustainable solutions that improve the quality of life and improve equality.

For example, access to clean water and sanitation is produced by engineers designing and implementing systems for water conservation and treatment globally. They often design efficient irrigation systems, and constructive passive water treatment plants